CSE6242

Predicting for Punctuality: Visualizing US Airline Delays

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1. Motivation

Flight delays are common but most prediction tools only focus on individual flights without considering the broader network context, such as how connected or central an airport sits in the network.

Why it matters?

Delays affects everyone - passengers, airports, airlines and even the economy. Highly connected hub airports face fundamentally different delay dynamics than smaller spokes. Understanding delay patterns across the network would help travelers plan around high-risk routes and times, and help airlines manage their operations more effectively.

2. Dataset

Source: U.S. Flight data (Kaggle)

Characteristics

- Over 7million rows
- Includes features like origin, destination, carrier, time and delays

Delay Definition

- A flight delay is formally defined as an arrival or departure occurring more than 10 minutes after the scheduled time

3. Approach

We built a tool that combines both machine learning with network analysis and interactive visualisation to predict and explore flight delays.

How it works?

- Predicting delay probability (classification)
- Predict delay magnitude (regression)
- Network graph of airports to capture network features (e.g. in/out degree)
- Visualisation through an interactive dashboard and simulation tool

Why this works?

- This approach captures both **individual flight factors and an airport's structural role in the network**
- Uses **tree-based models** to handle **complex non-linear relationships**
- **Combines prediction with visualisation** for interpretability and exploration
- Goes beyond **single-flight prediction by incorporating network-level context**

4. Models and Features

Network Features

To capture the structural role of airports and the inter-connectedness of the aviation network, we engineered features from the flight-level data to incorporate network topology metrics such as **node degree (traffic volume)**, **betweeness centrality (hub importance)**, and **edge weights (route frequency)** as part of our initial experiments.

Models

Tree-based ensemble models was used - **XGBoost, Catboost and Random Forest** as the underlying data contained categorical features such as the origin/destination airports, and also because we were careful not to assume normality in the distribution of the delay data. These models are also robust to outliers and skewed delays, able to handle categorical data and also surface features based on their predictive power.

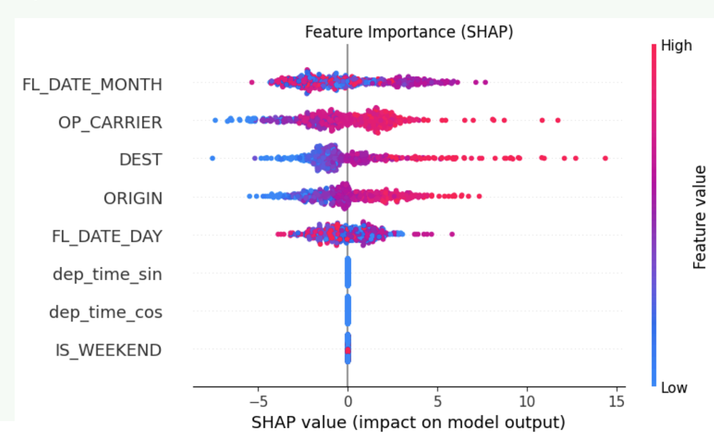
5. Modelling Results

Modelling for Delay Magnitude

Model	MAE	Accuracy	ROC/AUC
Linear Regression	20.77	-	-
XGBoost	20.40	0.80	0.74
CatBoost	-	0.79	0.69
RandomForest	20.46	0.79	0.66

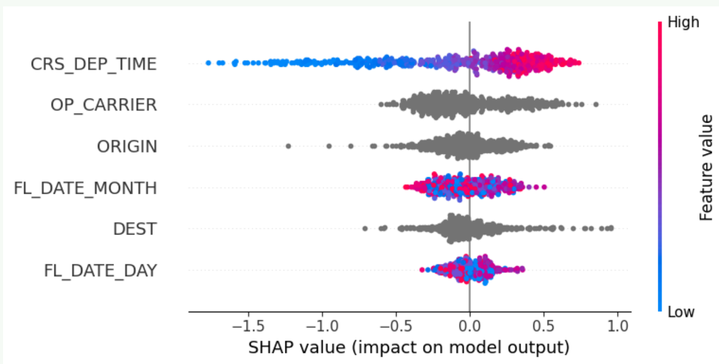
In modelling for the delay magnitude, XGBoost performed the best in terms of MAE. It remained the most performant in terms of accuracy and ROC/AUC as well.

Given XGBoost's performance, additional SHAP analysis indicates that **FL_DATE_MONTH** and **OP_CARRIER** as key drivers of delay magnitude, reflecting seasonal demand and airline operations.



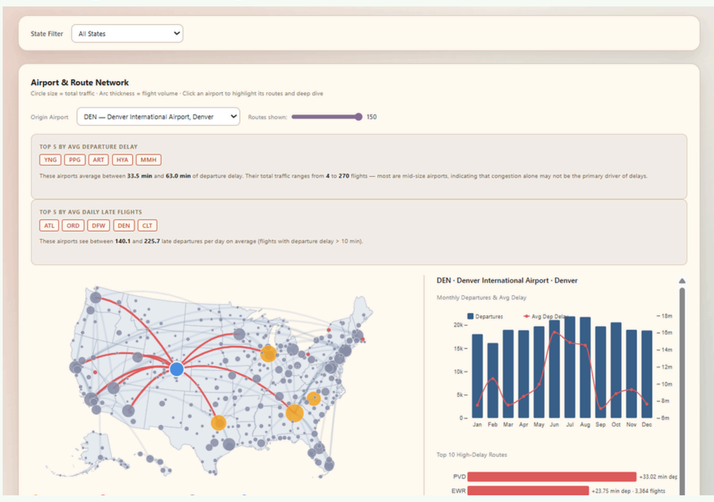
Modelling for Delay Probability

Similarly, when predicting the delay probability, similar feature emerge as strong predictors, such as the **scheduled flight time**, CRS_DEP_TIME, and **season factors capture** in FL_DATE_DAY and FL_DATE_MONTH.



6. Visualisations

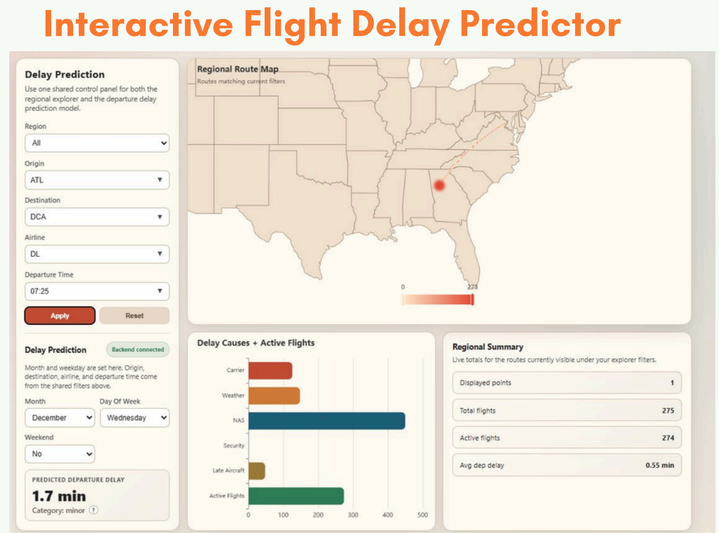
Flight Delay Interactive Dashbord



Dashboard explores delay patterns across routes, airlines and time. Enables state-level filtering to identify bottlenecks in a region.

Airport & Route Network - Identifies congested airports (hubs), highlights underperforming routes and compares daily late flights across airports.
Delay Heatmap - Shows delays across day of week x month and reveals peak delay periods
Carrier Comparsion - Ranks airlines by average delay depatures, highlights most vs least realiable carriers

Interactive Flight Delay Predictor



Predictor combines a shared control panel with a route map and prediction model. Users select flight details to see predicted delay duration and severity category.

Regional Route Map - Visualises the selected route on an interactive map, color-coded by delay volume to highlight high-traffic corridors.
Delay Causes & Active Flights - Breaks down delay causes (carrier, weather, NAS, security, late aircraft).
Regional Summary - Displays aggregate statistics for routes including total flights and average delay.

7. Conclusion

Key Takeaways

The results highlight the value of incorporating network-based features in improving predictive performance. By integrating predictive modelling with interactive visualization, the system provides users with both accurate predictions and intuitive tools to explore delay patterns across routes and airports

However, the study is limited by the use of single-year datasets and does not fully capture long-term trends or even extreme disruptions. Future work could then extend beyond this approach by incorporating real-time data.